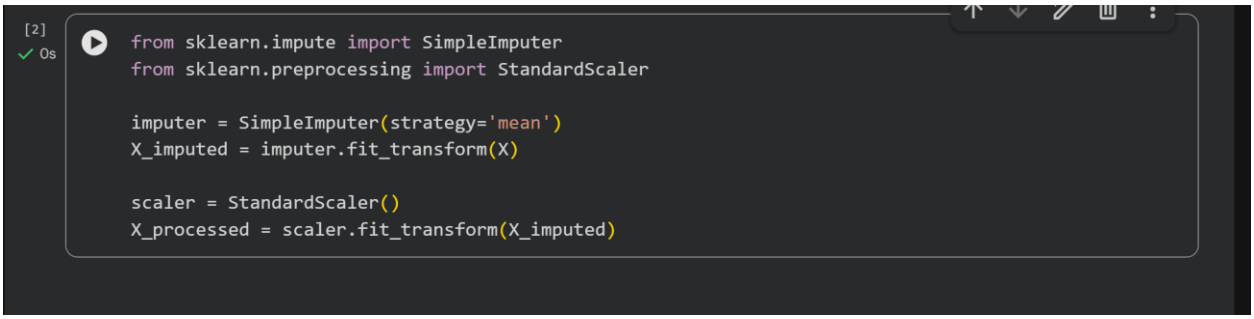
LAB # 05 RANDOM FOREST CLASSIFIER

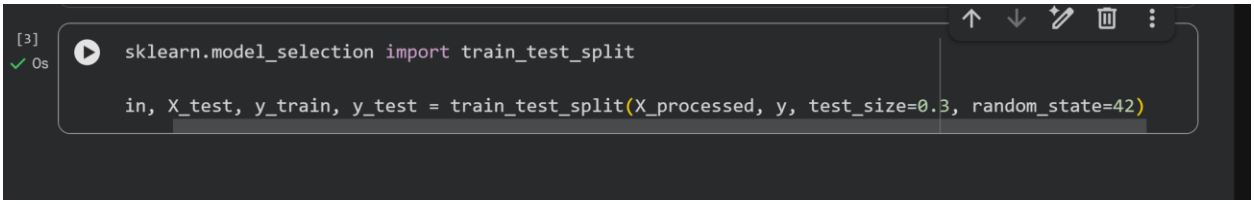
Task 1:



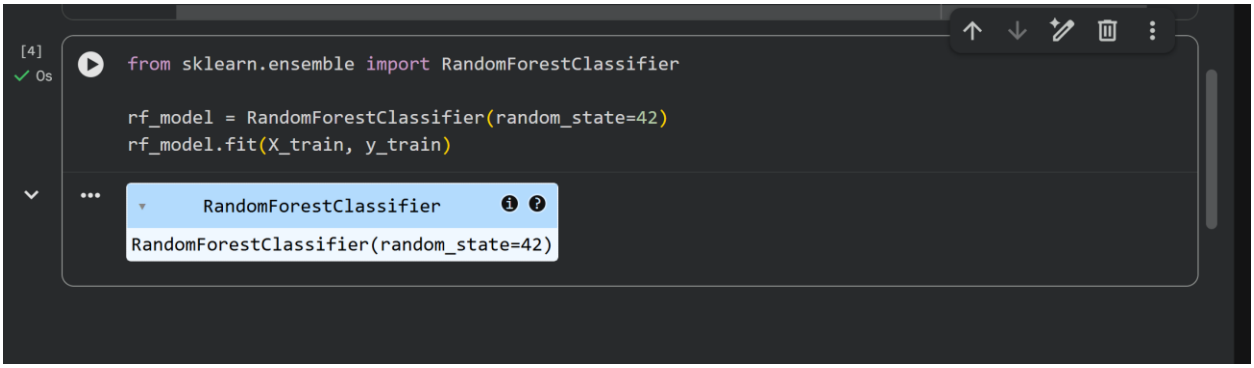
Task 2:



Task 3:



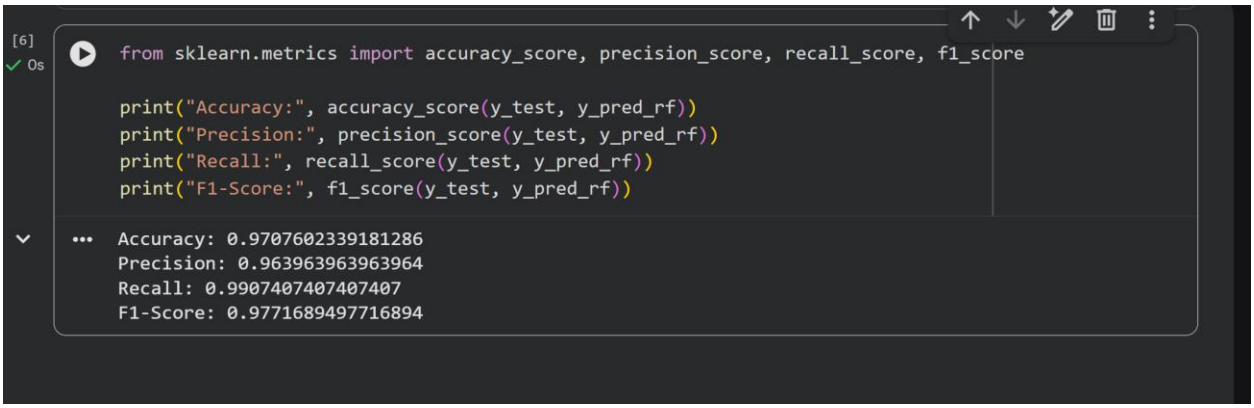
Task 4:



Task 5:



Task 6:

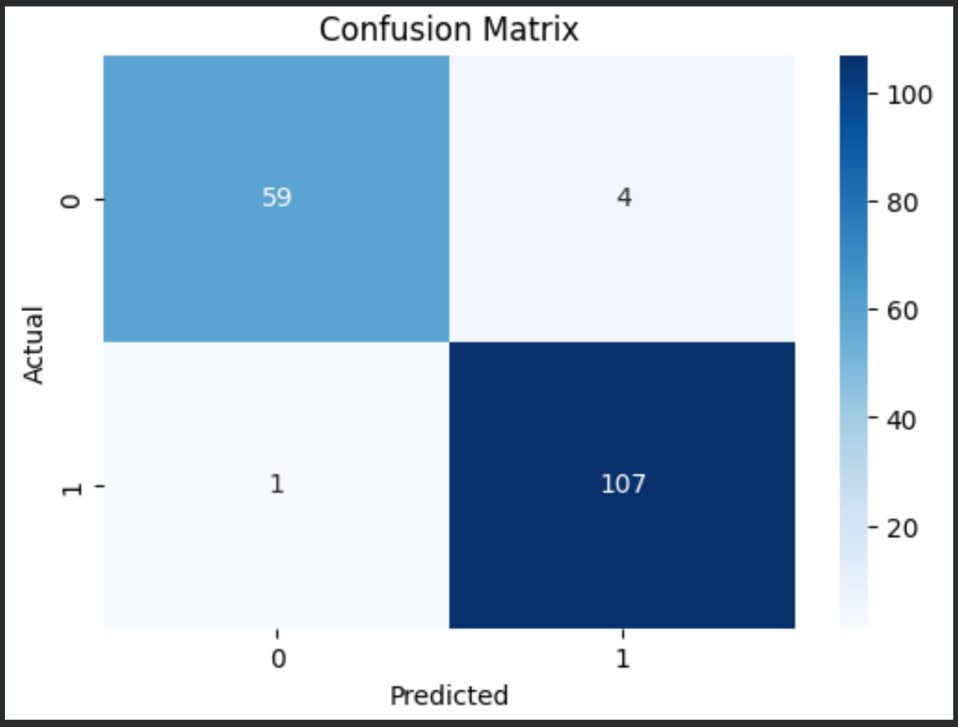


Task 7:

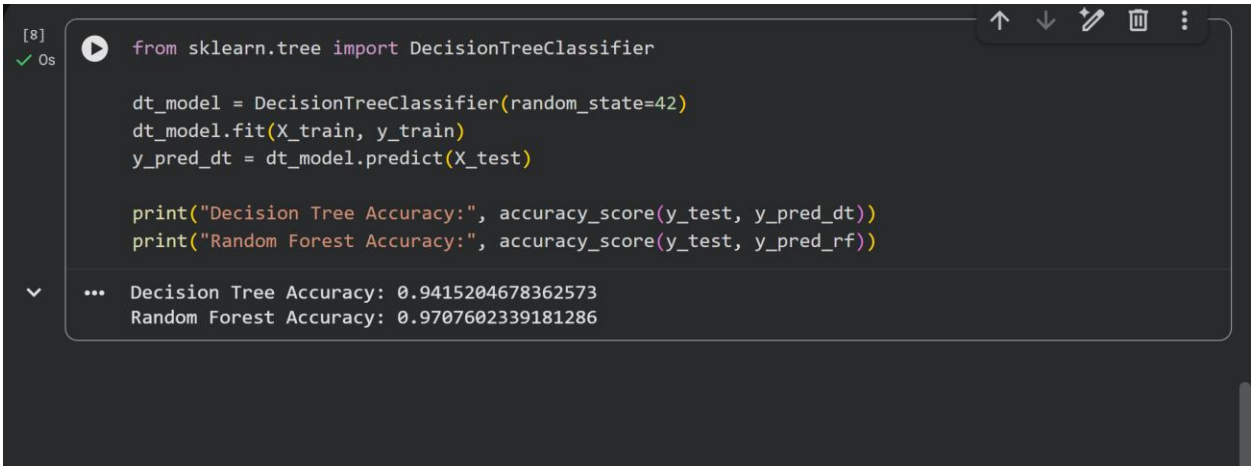
[7]
✓ 0s

```
import matplotlib.pyplot as plt
import seaborn as sns
from sklearn.metrics import confusion_matrix

cm = confusion_matrix(y_test, y_pred_rf)
plt.figure(figsize=(6, 4))
sns.heatmap(cm, annot=True, fmt='d', cmap='Blues')
plt.xlabel('Predicted')
plt.ylabel('Actual')
plt.title('Confusion Matrix')
plt.show()
```



Task 8:


[8]
0s
from sklearn.tree import DecisionTreeClassifier
dt_model = DecisionTreeClassifier(random_state=42)
dt_model.fit(X_train, y_train)
y_pred_dt = dt_model.predict(X_test)
print("Decision Tree Accuracy:", accuracy_score(y_test, y_pred_dt))
print("Random Forest Accuracy:", accuracy_score(y_test, y_pred_rf))
Decision Tree Accuracy: 0.9415204678362573
Random Forest Accuracy: 0.9707602339181286